

Wind Farm Project Finance - One-Page Brief

FIN-416 Commercial Bank Management | Project finance, credit analysis, and Excel modeling

Project snapshot

Dimension	Summary
Case	Hypothetical \$500M South Dakota wind power facility
Objective	Develop a viable financial plan that meets return and bank covenant targets
Capital structure	Senior debt, mezzanine capital, sponsor equity, and limited-recourse support
Primary outputs	ROE, free cash flow, ADSCR, LLCR, ICR, and scenario results

Key takeaways

- The base case produced a return above the required 25% threshold while satisfying the lender covenant package.
- Stress cases showed the project could remain feasible under lower MWh production and a later-period price decline.
- The analysis connects operating assumptions to lender risk through debt-service, reserve funding, and coverage-ratio constraints.

Selected metrics

Metric	Base case	Volume stress	Volume + price stress
ROE	29.1%	25.2%	25.0%
Minimum ADSCR	1.60+	1.60+	1.60+
Minimum LLCR	2.31+	2.30+	2.28+
Minimum ICR	1.76+	1.76+	1.76+
Covenants	Met	Met	Met

Portfolio signal

Demonstrates project-finance modeling, bank-style credit analysis, scenario testing, and financial communication for a capital-intensive energy project.